BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: KT-IX-23-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** Code: A

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. The study of the internal structure of the organism is .....

(A) morphology
(B) anatomy
(C) histology
(D) cytology

Qii. In specific method, collected data analyzed by using statistical method like .....

(A) addition only
(B) division only
(C) subtraction only
(D) ratio and proportion

Qiii. Which of the following organism belongs to kingdom Monera?

(A) Cyano bacteria
(B) Amoeba
(C) Mushroom
(D) Paramecium

Qiv. Nerve cells are long and are used for the ....

(A) transmission of nerve impulse
(B) transmission of blood
(C) production of carbohydrates
(D) Production of fats

Qv. A cell has ribosomes smaller in size and lack of membrane bounded nucleus. This cell represent to .....

(A) animals
(B) plants
(C) Prokaryote
(D) Eukaryote

Qvi. If process of meioses occurs in two parent cells. It will produce .....

(A) four haploid daughter cells
(B) four diploid daughter cells
(C) eight haploid daughter cells
(D) eight diploid daughter cells

Qvii. Attachment of chromosomes with spindle fibers and formation of an equatorial plate are the main events of .....

(A) Telophase-II
(B) Metaphase-II
(C) Prophase-II
(D) Anaphase-II

Qviii. Active sites of each enzyme have specific shape that fits with ...

(A) specific substrate
(B) specific complex
(C) specific product
(D) specific name

Qix. The site of Calvin cycle is .....

(A) Grana
(B) Stroma
(C) Mitochondria
(D) Golgi bodies

Qx. High concentration of carbon dioxide in photosynthesis ....

(A) increases the rate of dark reaction
(B) decreases the rate of dark reaction
(C) has no effect on dark reaction
(D) stop dark reaction

Qxi. A person has abdominal bloating. Which of the following is the best treatment of this symptoms of constipation?.

(A) taking more dietary fiber
(B) taking juices and food
(C) taking alkaline medicines
(D) taking spicy and acidic food

Qxii. The valve which prevents blood from flowing back into the ventricles are .....

(A) tricuspid valves
(B) bicuspid valves
(C) mitral valves
(D) semi lunar valves

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32

QQ.2. Attempt any EIGHT parts. All parts carry equal marks.

i Briefly explain the importance of fisheries.

ii Do you think viruses are parasites?. Justify your answers with examples.

iii Who proposed cell theory?. Also write its main points.

iv What are specific types of cells which make the xylem to perform conduction of water?.

v Enlist main events take place in S-phase of interphase.

vi Write events occur in the following phases of mitosis. (Table with columns: Phase | Write only one event; rows: Prophase, Metaphase, Anaphase, Telophase)

vii Why Lock and Key Model of enzyme action is important?

viii How Pyruvic acids goes through a series of reactions in Krebs cycle?

ix Discuss the role of dietary fiber in balanced diet.

x How do plants go through transpiration?

xi Briefly explain systemic circulation.

Section "C"

Maximum Marks: 21

Note: Note: Attempt any THREE of the following questions. All questions carry equal marks.

QQ.3.

a Briefly write the importance of proportion method in data analysis.

b Why species of Musk deer and Indus river blind dolphin are reducing in Pakistan?

QQ.4.

a Which cell organelles produce energy in prokaryotic cell?. How it is different from eukaryotic cell?.

b How mitotic apparatus is formed in animal and plants cells?.

QQ.5.

a Draw label diagram of Cofactors of enzymes.

b Explain Krebs cycle as an important step of aerobic respiration.

QQ.6.

a Differentiate between organic fertilizers and inorganic fertilizers with the help of suitable examples.

b Write one function of the following. (i). Superior vena cava (ii). Inferior vena cava (iii). Pulmonary vein (iv). Valves of vein.